Keeping What Matters

Predicting Which Post-Trained Behaviours Continual Training Erodes, Across Text and Image Inputs, and Aiming the Repair

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Abstract

Continual training on new domain data wears away the thin post-trained layer of a foundation model: safety, instruction following, tool use and the link between what the model sees and what it says. Production pipelines such as Thomson 1.0 repair the damage with replay, rehearsal and a merge chosen by sweeping. This proposal asks which post-trained behaviours continual training erodes, in what order, how fast and why, across text and image inputs, and how repair can be aimed at them. Pre-registered pilots on Qwen3.5 vision–language models show that erosion has an order (refusal, then skills, then knowledge), that a first-order forecast of erosion rates fails, that harmful photos are refused mostly through a route the text-derived refusal direction does not reach, and that replay triggered by an early warning used a fifth of the replay of a fixed schedule while ending 19.5 points ahead across four skills, in one run. The project comprises a behaviour-by-modality measurement of erosion with its early warnings and internal explanation, a repair controller that decides when, what and how much to repair, and a cross-modal ablation cost card for model releases.

1 Introduction

A foundation model is two things stacked: pretrained knowledge, deep and redundant, and a thin post-trained layer on top. That layer carries safety, instruction following, tool use and the link between what the model sees and what it says. Continual training wears it away, which is why Thomson 1.0 builds replay, a model merge and rehearsal into its pipeline [1]. Forgetting is known to follow laws [2], and benign fine-tuning to degrade safety [3, 4]. What is not known is which parts of the layer go first, whether their loss can be seen coming, and how to spend repair where it is needed rather than everywhere. Image inputs are almost unstudied here, although Thomson's base models are vision–language models and its continual pre-training measures no vision at all [1].

This proposal advances the thesis that post-training installs behaviours at different depths, and that continual training strips them from the outside in. Safety alignment is known to be shallow [5] and carried largely by a single direction [6]. Refusal and output-format conventions (short answers, valid tool calls) rest on few constraints and go first; abilities such as arithmetic go later, and pretrained knowledge goes last. The damage is front-loaded, concentrated while the learning rate is high. The hypothesis also explains the order from the inside: a behaviour resting on few constraints should be carried by few internal directions, which ablation can count. Image-input behaviour adds a modality axis, because part of it runs on routes that text-derived signals and repairs do not reach. The project addresses four research questions:

RQ1. Which post-trained behaviours does continual training erode, in what order, and does the number of internal directions that carry a behaviour predict when it erodes?

RQ2. Can a signal read during training warn before a behaviour is lost, where a forecast from the training data fails?

RQ3. Can repair aimed by such signals, deciding when, what and how much to replay or merge, keep more behaviour than fixed replay or merging at lower cost?

RQ4. Do text-derived signals and repairs reach image-input behaviour, and what does keeping native vision cost?

2 Related Work

Forgetting during fine-tuning follows predictable laws [2], and benign fine-tuning degrades safety, some data far more than other [3, 4]. CapTrack, the Lab's framework, tracks forgetting across capability types and is the measuring instrument this project extends with a modality axis [7]. These results establish that post-trained behaviour erodes. They do not establish which behaviours go first under whole-corpus continual training, as rates that can be forecast, whether a signal fires before they go, or how image inputs differ.

On repair, replay limits forgetting [8], as do progress & compress and functional regularisation [9, 10]. In continual pre-training, replay combined with learning-rate re-warming and re-decay matches full re-training, and the schedule itself shapes forgetting [11]; weight interpolation trades new learning against old [12], and fine-tuning on the model's own responses reduces forgetting [13]. Thomson uses replay and chooses its merge ratio by sweeping sixteen runs and ten ratios [1], and ADMIRE-BayesOpt searches data mixtures offline [14]. None of these decides, during training, when to repair, which behaviour, what dose, and at what cost to other behaviours.

For images, safety transfers poorly when a vision–language model is first built [15–18], and harmful narrow fine-tuning erodes vision–language safety more than text-only evaluations show [19]. The literature disagrees on a mechanism that decides cost. One view is that alignment learned from text covers image inputs through the shared language model [20–22]; the other is that image inputs fall outside it [17–19]. For real photos the pilots side with the second (Section 4).

What continual training does to image-input behaviour afterwards, the trajectory rather than the starting point, has not been measured; if text-side repair does not reach images, keeping native document vision has a price, and this project measures it.

3 Proposed Research

The programme has three parts, each using the outputs of the one before and each ending in written-up work. Part 1 builds the measurement, the warning signals and the internal explanation; Part 2 uses them to aim the repair; Part 3 turns the cross-modal measurements into a standard for model releases. Experiments use the Qwen3.5 vision–language ladder, the family Thomson builds on (Section 5).

3.1 Measure and predict (Part 1, months 0–12; RQ1, RQ2)

A behaviour-by-modality grid extends CapTrack's CAN, WILL and HOW axes [7] to image inputs and scores output format and ability separately. Generation-free reads are validated against generated responses, including a judge for image responses, which the pilots lack. Trajectories are measured across the ladder (0.8B–9B) and all four corpora, with training strength reported as learning rate × optimiser steps plus batch size, because with Adam, learning rate × tokens is not comparable across batch sizes. For the explanatory half of RQ1, the directions each behaviour needs are counted by ablation, for each behaviour and modality, and tested against the order of erosion: behaviours carried by few directions (refusal, output formats) should erode first. Because the starting gradient cannot rank corpora (Section 4), RQ2 is addressed with signals read during training, a refusal canary and each behaviour's loss on the model's own answers, together with locating when damage concentrates, including the effect of learning-rate re-warming [11].

3.2 Aim the repair (Part 2, months 12–24; RQ3, RQ4)

Part 2 opens by replicating the on-demand replay result (Section 4) across data orders and at 2B, and by testing the timing hypothesis directly with identical bursts early and late. It then builds a repair controller that decides when (warning signals), what (behaviour family; lost format or lost ability) and how much, since fixed replay has side effects. At matched compute, the controller is compared with fixed replay [8], self-distillation [13], functional regularisation [9, 10] and merging [12]. The merge ratio is derived from predicted end states instead of being swept [1]; ADMIRE-BayesOpt is the offline counterpart [14]. For RQ4, Part 2 tests whether text-only repair protects image inputs and finds the minimum image replay that does. The output is the frontier of protection against replay and compute spent, including what native document vision costs to keep.

3.3 Harden and standardise (Part 3, months 24–33; RQ4)

Part 3 turns the cross-modal ablation results into a cost card for model releases: what stripping safety or values costs, per modality, complementing dual-modal defences on the attack side [23]. It also tracks constitutional adherence through every pipeline stage for both modalities, and tests whether objectives with more independent constraints spread behaviours across more directions (P4, suggestive so far).

4 Preliminary Results

All pilots used Qwen3.5-0.8B, with 2B for checks, in a gentle regime (SGD, 16-bit) and a realistic one (AdamW with full-precision master weights, learning rates 2e-6 to 2e-5, 16k-token batches); data and measures are described in Section 5. Refusal is read without generation, as the first-token log-odds gap between matched harmful and harmless requests, and is validated against generated responses for text inputs; the 476 skill items are scored individually so that arms can be compared item by item. The pass/fail rule of every prediction was registered before its run (PR), and Table 1 summarises where each prediction stands.

Table 1: Predictions and their status. "Pre-registered" means the pass/fail rule was fixed before the run (PR); failures are kept.

Prediction (ID)	Claim	Status and evidence
Order (P1)	Post-trained behaviour erodes before knowledge; safety before skills.	**Supported, pre-registered** (4 of 4 checks). Knowledge goes last but is not immune on chat-like data (nb03, nb06).
Modality gap (P2)	Image-input behaviour erodes faster than text-input behaviour.	**Qualified:** 2× at 0.8B (two seeds, first-token measure), not at 2B, not yet in generated responses (nb04).
Shared route (P3)	Text-derived repair covers image inputs.	**Static:** the route is shared for rendered text (62–68%), mostly not for photos (34%). **Dynamic: refuted;** text-only replay moved image refusal unpredictably (nb03, nb05).
Few directions (P4)	Behaviours shaped by fewer constraints are carried by fewer internal directions.	**Suggestive:** one direction removes almost all text refusal but about two thirds of value behaviour; the link to erosion order is untested (nb05, nb07).
Forecastable rate (P5)	A score computed before training predicts each corpus's erosion rate.	**Not supported:** it fits the known rates (Spearman 0.94), but 1 of 4 forecasts pass (nb04).

Early warning (R3)	A cheap signal crosses before any skill falls.	**Supported, pre-registered:** refusal canary at 131k tokens, first skill loss at 262k (one run, nb06).
Aimed repair (R2)	On-demand replay matches fixed replay at half the replay or less.	**Supported, pre-registered, and exceeded:** 19% of the replay, +19.5 points (one run, nb06).

Erosion has an order, and skills erode in two ways. At the gentlest realistic rate, refusal lost 5.6% (code) and 17% (chat-like) by 262k tokens, before any skill moved, and all four pre-registered order checks passed (nb03). On chat-like data at lr 5e-6, refusal fell 12% by 131k tokens, tool calling and instruction following at 262k, and knowledge only from 1M; on code, knowledge stayed within 8 of 300 items over 4M tokens. The kind of text matters: chat-like text erodes refusal 2.5–6× faster than code, with legal and news prose in between (about 6% per million tokens). Skills fail in two ways. Chart reading and tool calling lose their output format (short answers fall from 0.98 to 0.04–0.07; valid calls from 0.86 to 0.16–0.20), while maths keeps its format and loses accuracy.

Rates could not be forecast from the starting gradient. A first-order score measures how well a corpus's mean gradient aligns with the gradient of the refusal gap on the untrained model. It ranked the six known rates almost perfectly (Spearman 0.94) but passed only 1 of 4 pre-registered forecasts for legal and news text (nb04). It separates dangerous corpora from mild ones but cannot rank the mild ones, which is why Part 1 turns to early-warning signals.

Image inputs and internal routes. At 0.8B, first-token image refusal eroded twice as fast as text refusal on chat-like data, replicated across seeds; it did not at 2B, and it does not yet show in generated responses (nb04). Ablation, which removes a behaviour's direction and measures what is left, gives four results. Photos have their own refusal route: removing the text-derived refusal direction leaves 32–38% of refusal for requests rendered as images and 66% for harmful photos; the photo direction sits 78° from the text one, and removing it takes away 89% of photo refusal but only a third of text refusal (2B; nb05). Text-derived monitoring and repair therefore see only part of image-input behaviour. Values have a picture-only route too: a third of image-input value behaviour can be removed at no cost to ability, while removing it with the shared part costs 15 ARC points (nb05), the price curve Part 3's cost card would report. Geometry misleads: Qwen3.5-2B's refusal subspace looks spread out (stable rank 8.4), yet one ablation removes 68.5% of refusal (nb07), so every measure here is functional. And an alarm on the raw refusal level could not be calibrated, because the baseline drifts by 4 SD (nb07); the canary below reads the drop from the run's own starting level instead.

Aiming the repair. A replay pilot showed that repair is not neutral: replaying the model's own chat answers restored refusal but wiped out tool calling (0.78 → 0.00–0.08; nb03). W2 then compared five arms on chat-like data (lr 5e-6, 2M tokens, one data order; PR §7): no replay; fixed replay of the model's own answers across five behaviour families; the same replay on demand, triggered by a one-shot refusal warning and per-family drift sensors calibrated on the no-replay arm by a fixed rule; fixed replay of reference answers; and fixed replay of chat only, plus a merge sweep. Three of six pre-registered predictions passed (R2, R3 and R5); R1 and R6 failed on chart reading, and R4 on tool calling (Figure 1, Table 2). Two bursts, at 131k and 459k tokens, were enough: with 19% of the fixed schedule's replay, on-demand replay ended 19.5 points ahead on all 476 items (95% CI 14.5 to 24.4), at baseline on instruction following and maths, with all of the domain learning kept. Fixed replay protected tools but lost chart reading and cost 21 maths points against no replay; reference answers moved skills 17 points further than the model's own; chat-only replay protected tools 41 points less than broad replay; and merging at $\alpha = 0.5$ restored chart reading to 99% of its starting level but tools only to 77%. The refusal warning fired one checkpoint before the first skill fell.

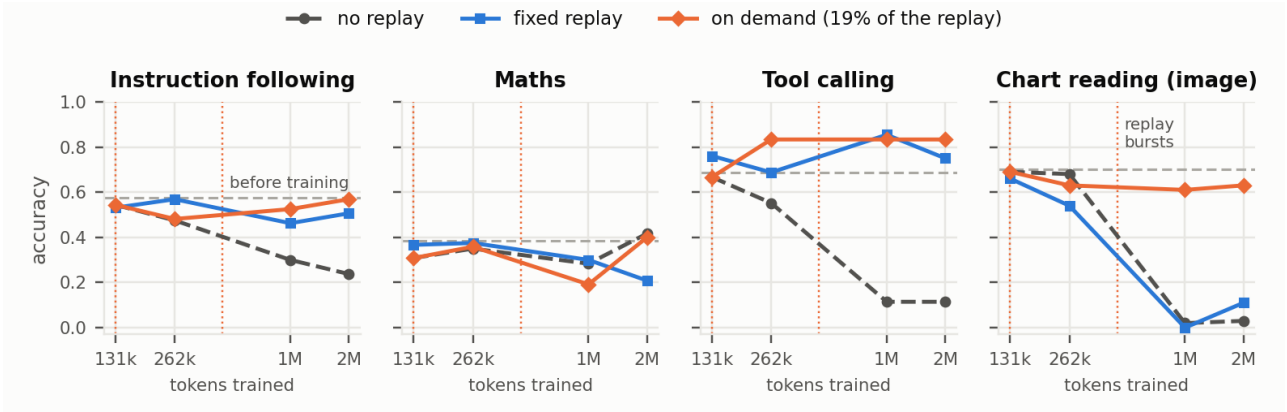


Figure 1: Skills over 2M tokens of chat-like continual training (Qwen3.5-0.8B). Dashed grey: accuracy before training. Dotted lines: the two on-demand replay bursts. On-demand replay ends near its starting level on all four skills; fixed replay keeps tools and instruction following but loses maths and chart reading (nb06).

Table 2: W2 at 2M tokens. Accuracy on 160 / 120 / 96 / 100 items. Domain learning kept is the share of the no-replay arm's fall in held-out chat loss. Tool scores above the starting level are partly template learning, because the replay pool and the test share templates.

W2 arm, at 2M tokens	Replay (share of fixed)	Instruction	Maths	Tools	Charts	Domain learning kept
Before training	–	0.58	0.38	0.69	0.70	–

No replay	0	0.24	0.42	0.11	0.03	100%
Fixed broad replay	100%	0.51	0.21	0.75	0.11	91%
On demand	**19%**	**0.57**	**0.40**	**0.83**	**0.63**	**100%**
Fixed, reference answers	87%	0.28	0.05	0.95	0.00	92%
Fixed, chat only	94%	0.31	0.12	0.34	0.00	96%
Merge, $\alpha = 0.5$	0	0.48	0.21	0.53	0.69	84%

Two cautions set how far this goes. The on-demand arm matched the no-replay arm exactly until its first burst, so the difference is caused by the repair, but it is one run of one small model, so replication across data orders and sizes opens Part 2. And what worked was timing and dose, not routing: the per-behaviour sensors triggered replay once. Timing also points to the front-loading hypothesis. Without replay, the loss on the model's own answers rose mostly between 131k and 507k tokens, around the learning-rate peak; after the bursts it stayed flat.

5 Models, Data and Evaluation

Models are the Qwen3.5 vision–language ladder: 0.8B and 2B for development, as in the pilots, and 4B and 9B on Lab compute for thesis-grade runs at production-like batch sizes. Training corpora are code, chat-like text, US case law and news, the last two standing in for Thomson's domain. Refusal uses matched harmful and harmless requests in plain, wrapped and image-rendered framings, with natural photos from VLSBench as an external check; skills use instruction following (160 items), maths (120), tool calling (96) and chart reading on images (100); knowledge uses ARC, as text and rendered as images. Arms are compared with paired tests over the same items, and every prediction is registered with its pass/fail rule before the run that tests it.

6 Risks and Mitigation

The principal risk is that the on-demand result, from one run, does not replicate. Part 2 opens with its replication across data orders and at 2B, and the measured costs of fixed repair (Table 2) motivate aimed repair even if the advantage shrinks. Small-model results may not transfer, so the ladder extends to 9B at production-like batch sizes, with results reported within explicit scale bounds. Image refusal is read from the first token and not yet validated over training; a judge for generated image responses is built in Part 1, and until then image claims stay at the level of inclination. Warning signals may give too little lead time, so several are tested across learning rates, with protection through the high-rate window as the fallback. The tool test shares templates with the replay pool, so held-out templates are added and tools are reported as kept, not improved. If the working hypothesis fails, Parts 1 and 2 do not depend on it, and the work continues as an empirical programme on when and how to repair.

7 Expected Contributions

The project is expected to make three contributions, each targeting a peer-reviewed paper (Table 3):

C1. A behaviour-by-modality account of erosion under continual training: its order, the early-warning signals that precede it, and the internal directions that explain it (Part 1; ICLR or NeurIPS).

C2. A repair controller that decides when, what and how much to replay or merge, with the frontier of protection against compute spent and the cost of keeping native vision (Part 2; ICML, NeurIPS or CoLLAs).

C3. A cross-modal ablation cost card for model releases, with constitutional adherence tracked through training (Part 3; NeurIPS Datasets and Benchmarks).

Practically, these would tell a pipeline like Thomson's when and where to spend repair, instead of sweeping it. The work builds on the Lab's own methods: Thomson 1.0 is the setting [1], CapTrack the instrument [7], experience replay the repair that W2 already aims [8], and progress & compress, functional regularisation and ADMIRE-BayesOpt the comparisons [9, 10, 14]. The accompanying repository registers every experiment against a claim before it runs and keeps its failures.

8 Timeline and Outputs

Table 3: Work plan in order of dependency; the core alone is a complete thesis.

Months	Core	Extension	Written output
0–12	Part 1: measurement grid; trajectories across the ladder; warning signals; directions per behaviour	4B and 9B on Lab compute	Literature review; Part 1 paper (ICLR / NeurIPS)
12–24	Part 2: W2 replication; early vs late bursts; repair controller against fixed replay, self-distillation, regularisation and merging	Minimum image replay	Part 2 paper (ICML / NeurIPS / CoLLAs)
24–33	Part 3: cross-modal ablation cost card; constitutional adherence through training	Constraint-diverse objectives (P4)	Part 3 paper submitted (NeurIPS Datasets and Benchmarks)

33–36	Thesis writing	–	Thesis submitted

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